Reward Recovery in Sequential Decision Models

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1 Setup and Notation

Consider a stationary infinite-horizon MDP with finite state space $\mathcal{S}$, finite action space $\mathcal{A}$, discount factor $\beta \in (0,1)$, transition kernel $P : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})$, and unknown reward $r : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$. Additive T1EV shocks yield the soft Bellman recursion.

$$Q^*(s,a) = r(s,a) + \beta \sum_{s'} P(s' \mid s,a)\, V^*(s'), \tag{1}$$

$$V^*(s) = \log \sum_{a'} \exp Q^*(s,a'), \tag{2}$$

$$\pi^*(a \mid s) = \exp\big(Q^*(s,a) - V^*(s)\big), \tag{3}$$

$$A^*(s,a) = Q^*(s,a) - V^*(s) = \log \pi^*(a \mid s). \tag{4}$$

The advantage A^* is identified from CCPs. The reward r is not: any $\tilde{r}(s,a) = r(s,a) + \lambda(s)$ generates identical CCPs (Magnac and Thesmar, 2002). Following Kalouptsidi et al. (2021), call the set of all such $\tilde{r}$ the identified set.

2 Four Approaches to Reward Recovery

2.1 Reduced-Form Q

The reduced-form approach estimates $Q_\omega(s,a)$ by logit MLE.

$$\hat{\omega} = \arg\max_{\omega} \sum_{i,t} \log \frac{\exp Q_\omega(s_{it}, a_{it})}{\sum_{a'} \exp Q_\omega(s_{it}, a')}. \tag{5}$$

At the population optimum, $Q_{\hat{\omega}}(s,a) = Q^*(s,a) + c(s)$ for some action-independent c, so the backed-out "reward" is the advantage $A^*(s,a)$, not the true reward $r(s,a)$.

2.2 AIRL (Fu et al., 2018)

The AIRL discriminator decomposes as follows.

$$f_\theta(s,a,s') = g_\theta(s,a) + \beta\, h_\phi(s') - h_\phi(s), \tag{6}$$

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Here g_θ approximates the reward and $\beta h(s') - h(s)$ is potential-based shaping (Ng et al., 1999). At the GAN optimum the discriminator score equals the advantage, $f^* = A^*$ (Fu et al., 2018, Appendix A.4), which implies the following expression for the recovered reward network.

$$g^*(s,a) = r(s,a) - \delta(s) + \beta \sum_{s'} P(s' \mid s,a)\, \delta(s'), \tag{7}$$

where $\delta(s) \equiv V^*(s) - h^*(s)$. Unless $\delta \equiv 0$, the recovered g^* is a potential-based perturbation of r.

2.2.1 LSW Anchor Identification (Lee et al., 2026)

LSW impose two anchors. First, the exit payoff is normalized to zero, so $g(s, \text{exit}) = 0$ for all s. Second, the shaping function is pinned at the absorbing state, so $h(s_{\text{abs}}) = 0$, where exit deterministically transitions to s_{abs}.

Lemma 1 (Non-identifiability)**.** *If (g,h) and $(\bar{g},\bar{h})$ produce the same f for all (s,a,s'), then $\bar{h} = h + \Phi$ and $\bar{g}(s,a) = g(s,a) + \Phi(s) - \beta\Phi(s')$ for some $\Phi : \mathcal{S} \to \mathbb{R}$.*

Proof. Rearrange $g + \beta h(s') - h(s) = \bar{g} + \beta\bar{h}(s') - \bar{h}(s)$ and define $\Phi = \bar{h} - h$. □ □

Theorem 2 (Anchor identification)**.** *Under (A1)–(A2), if (g,h) and $(\bar{g},\bar{h})$ realize the same f, then $g = \bar{g}$ and $h = \bar{h}$.*

Proof. By Lemma 1, $\bar{g}(s, \text{exit}) = 0 + \Phi(s) - \beta\Phi(s_{\text{abs}})$. Anchor (A1) forces $\Phi(s) = \beta\Phi(s_{\text{abs}})$ for all s. Anchor (A2) forces $\Phi(s_{\text{abs}}) = 0$. Hence $\Phi \equiv 0$. □ □

Theorem 3 (Recovery of true reward)**.** *If the true pair (r^*, V^*) satisfies $r^*(s, \text{exit}) = 0$ and $V^*(s_{\text{abs}}) = 0$, then any AIRL pair (g,h) realizing A^* under the same anchors satisfies $g = r^*$ and $h = V^*$.*

Proof. Define $\Phi = h - V^*$. The exit anchor gives $\Phi(s) = \beta\Phi(s_{\text{abs}})$. The absorbing anchor applied to both h and V^* gives $\Phi(s_{\text{abs}}) = 0$. □ □

The Kang et al. (2025) ERM framework uses the same anchor principle (fixing the reward of a known action) but solves a single non-adversarial optimization.

2.3 IQ-Learn (Garg et al., 2021)

The inverse Bellman operator $r_Q(s,a) \equiv Q(s,a) - \beta \sum_{s'} P(s' \mid s,a)\, V_Q(s')$ expresses reward as a function of Q alone. IQ-Learn estimates Q by solving the following optimization problem.

$$\max_Q \; \mathbb{E}_{(s,a)\sim\rho^*}\big[\phi\big(r_Q(s,a)\big)\big] - (1-\beta)\, \mathbb{E}_{s_0\sim\mu_0}\big[V_Q(s_0)\big], \tag{8}$$

The function $\phi(x) = x - x^2/2$ corresponds to χ^2 divergence. At the optimum, r_Q separates immediate reward from continuation values via the Bellman constraint.

2.4 GLADIUS (Kang et al., 2025)

GLADIUS parameterizes both $Q(s,a)$ and $\hat{\mathbb{E}}[V(s') \mid s,a]$ with neural networks, trained jointly via logit NLL plus a Bellman consistency penalty on observed (s,a,s') transitions. It extracts reward via the same inverse Bellman operator as IQ-Learn. The MLP architecture shares parameters across states, providing implicit regularization over tabular Q.

3 Counterfactual Taxonomy

Definition 4 (Type I: state extrapolation)**.** *The transition kernel P and reward r are unchanged. Only the realized state values shift. The counterfactual CCPs require only evaluating the estimated Q at new state inputs, with no Bellman re-solve.*

Definition 5 (Type II: environment change)**.** *The reward r is unchanged, but the transition kernel shifts to $\tilde{P}$ or the action set changes. The counterfactual Q must satisfy a new Bellman equation $\tilde{Q} = r + \beta\tilde{P}\tilde{V}$, which requires the structural reward r separated from the old continuation values.*

Proposition 6 (Type I equivalence)**.** *Under Type I, all four methods yield identical counterfactual CCPs.*

Proof. Each recovers $Q^*(s, a) + c(s)$ for some action-independent c. The softmax $\pi(a \mid \tilde{s}) \propto \exp(Q^*(\tilde{s}, a) + c(\tilde{s}))$ cancels c. □ □

Proposition 7 (Type II divergence)**.** *Under Type II, reduced-form produces invalid predictions. AIRL with anchors, IQ-Learn, and GLADIUS produce valid predictions when r is correctly recovered.*

The reduced-form estimate $Q^* + c(s)$ bundles r with βPV^*. Under $\tilde{P}$, continuation values change but the reduced-form has no way to disentangle r. AIRL with anchors recovers $g^* = r^*$ exactly (Theorem 3). IQ-Learn and GLADIUS separate r via the inverse Bellman operator. The LSW anchors provide what Kalouptsidi et al. (2021) call the "additional restrictions" needed to identify counterfactuals under transition changes.

4 Application to Lee, Sudhir & Wang (2026)

LSW study sequential content consumption with three actions (buy, wait, exit) using AIRL under anchors (A1)–(A2) and deterministic transitions. Their wait-time validation (Section 6.3) is a Type I counterfactual, so a reduced-form logit would pass this test identically. Their WFF removal and segment-specific delay policies (Sections 7.1–7.2) are Type II because they change the action-set structure and the MDP that each segment faces. Their content-based paywall (Section 7.3) is also Type II because high-content episodes become paid-only, altering the transition structure episode-by-episode.

5 Simulation Study

We construct a tabular MDP that mimics the LSW book-reading environment. The state encodes episode position, content quality, and wait-time level, giving $15 \times 3 \times 4 = 180$ regular states plus one absorbing state. The three actions are buy, wait, and exit, with deterministic transitions and $\beta = 0.95$. The true reward is $r^*(s, \text{buy}) = u(c) - 2.0$, $r^*(s, \text{wait}) = u(c) - 0.08w$, and $r^*(s, \text{exit}) = 0$, where $u \in \{0.5, 1.5, 3.0\}$. Analyses A through C and E through F are evaluated at the population level with no estimation noise. Analysis D uses finite-sample estimation.

5.1 A. Reward recovery

5.2 B. Type II error vs. transition shift

The counterfactual changes the transition kernel so that buying advances k episodes instead of one. The Bellman equation is re-solved under each method's recovered reward.

Method	MSE vs. r^*	Corr. with r^*
Oracle / AIRL+anchors / IQ / GLADIUS	0.000	1.000
AIRL-no-anchors	10.005	0.440
Reduced-form	40.020	0.138

Table 1: Population-level reward recovery. Type I CCP errors are below 10^{-6} for all methods.

k	Oracle/AIRL+a/IQ/GLADIUS	AIRL-no-a	Reduced-form
1	0.000	0.000	0.000
2	0.000	0.025	0.082
3	0.000	0.017	0.094
5	0.000	0.015	0.123
7	0.000	0.016	0.129
10	0.000	0.014	0.139

Table 2: Type II CCP error. Structural methods are exact. Reduced-form error grows with shift magnitude.

5.3 C. Shaping magnitude sweep

We construct shaped rewards $\tilde{r} = r^* + \alpha V^*(s) - \alpha\beta V^*(s')$ for $\alpha \in [0, 1]$ and evaluate the Type II counterfactual with $k = 3$.

α	Type II CF error	Corr. with r^*
0.00	0.000	1.000
0.10	0.004	0.908
0.30	0.011	0.610
0.50	0.017	0.440
1.00	0.027	0.270

Table 3: CF error is monotone in shaping magnitude. Without anchors, α is indeterminate.

5.4 D. Finite-sample convergence

The tabular χ^2 IQ-Learn objective has multiple local optima in finite samples. We compare three finite-sample estimators. The reduced-form logit converges to A^*, which is the wrong target for Type II counterfactuals. The anchor-constrained MLE adds soft penalties on the implied reward at exit and absorbing states to the logit NLL. GLADIUS uses neural Q and EV networks.

5.5 E. Anchor misspecification

We set the true exit payoff to ε while the anchors assume it is zero.

5.6 F. Content-based paywall

High-content episodes become paid-only and the wait action is blocked, forcing buy-or-exit. This is a Type II shift that mirrors Section 7.3 of LSW.

N	Reduced-form	Anchor-MLE	GLADIUS
500	0.264	0.255	0.106
2,000	0.262	0.255	0.123
10,000	0.262	0.257	0.099

Table 4: Type II CF error ($k = 3$, 2 seeds). GLADIUS cuts error by 60% over tabular methods via implicit neural regularization.

ε	Corr. with r^*	Type II CF error	In-sample π shift
0.0	1.000	0.000	0.000
0.5	0.976	0.002	0.000
1.0	0.909	0.004	0.000
2.0	0.729	0.008	0.000

Table 5: Degradation is smooth, not knife-edge. In-sample CCPs are unaffected.

6 Summary

References

Fu, J., Luo, K., and Levine, S. (2018). "Learning Robust Rewards with Adversarial Inverse Reinforcement Learning." ICLR.

Garg, D., Chakraborty, S., Cundy, C., Song, J., and Ermon, S. (2021). "IQ-Learn: Inverse soft-Q Learning for Imitation." NeurIPS.

Kalouptsidi, M., Scott, P. T., and Souza-Rodrigues, E. (2021). "Identification of Counterfactuals in Dynamic Discrete Choice Models." Quantitative Economics, 12(2), 351–403.

Kang, E. H., Yoganarasimhan, H., and Jain, L. (2025). "An Empirical Risk Minimization Approach for Offline Inverse RL and Dynamic Discrete Choice Models." arXiv:2502.14131.

Lee, P., Sudhir, K., and Wang, T. (2026). "Consumer Engagement with Sequential Content: A Content-Aware Dynamic Choice Model." Working paper, Yale University.

Magnac, T. and Thesmar, D. (2002). "Identifying Dynamic Discrete Decision Processes." Econometrica, 70(2), 801–816.

Ng, A. Y., Haley, D., and Russell, S. J. (1999). "Policy Invariance Under Reward Transformations." ICML.

Rust, J. (1987). "Optimal Replacement of GMC Bus Engines." Econometrica, 55(5), 999–1033.

Method	CCP error
Oracle / AIRL+anchors / IQ-Learn	0.000
AIRL-no-anchors	0.009
Reduced-form	0.185

Table 6: Reduced-form fails at 18.5pp because it cannot separate content utility from continuation value.

	Reduced-form	AIRL (generic)	AIRL (anchors)	IQ-Learn / GLADIUS
Recovers	$A^*(s, a)$	r + shaping	$r^*(s, a)$	$r^*(s, a)$
Type I CF	✓	✓	✓	✓
Type II CF	×	Only if $\delta \equiv 0$	✓	✓